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(54) **BATTERY MODULE AND BATTERY PACK INCLUDING THE SAME**

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(57) **ABSTRACT**

A battery module including: a battery cell stack in which a plurality of battery cells including electrode leads protruding in mutually opposite directions are stacked; a housing that houses the battery cell stack; a first heat sink and a second heat sink that are located under the bottom part of the housing, and first and second refrigerant flow paths formed between the first heat sink and the bottom part of the housing and between the second heat sink and the bottom part of the housing, respectively. The refrigerant flow path formed by the first heat sink and the refrigerant flow path formed by the second heat sink are separated from each other.

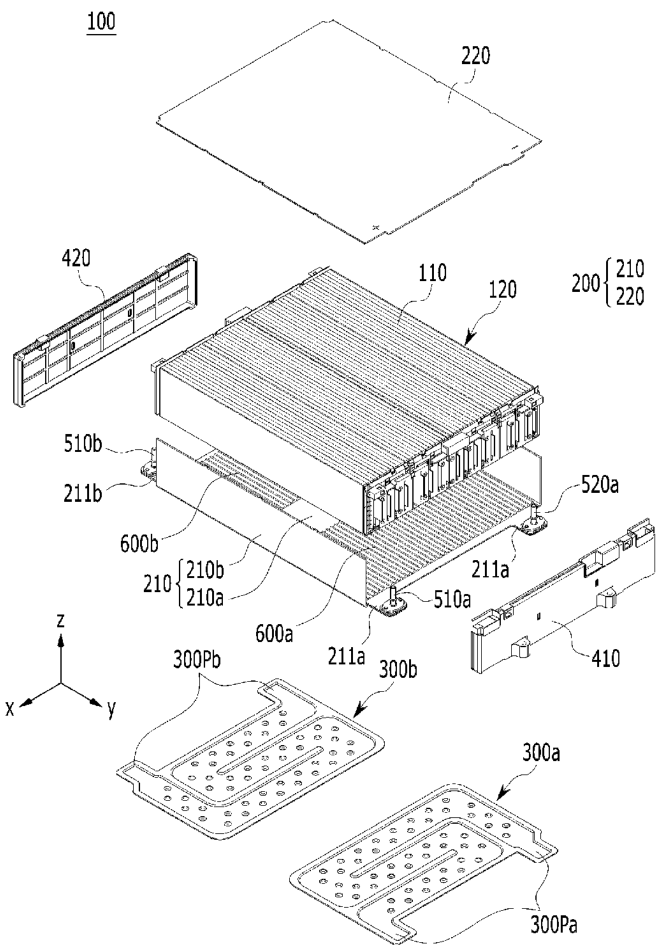


FIG. 1

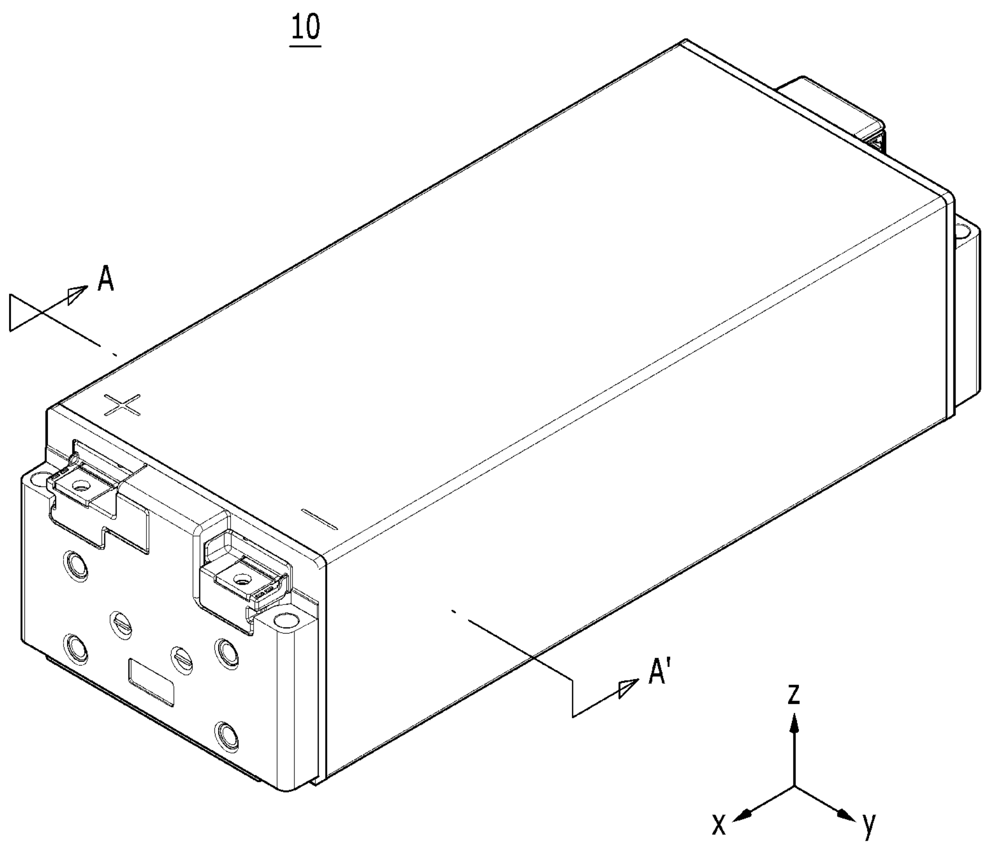


FIG. 2

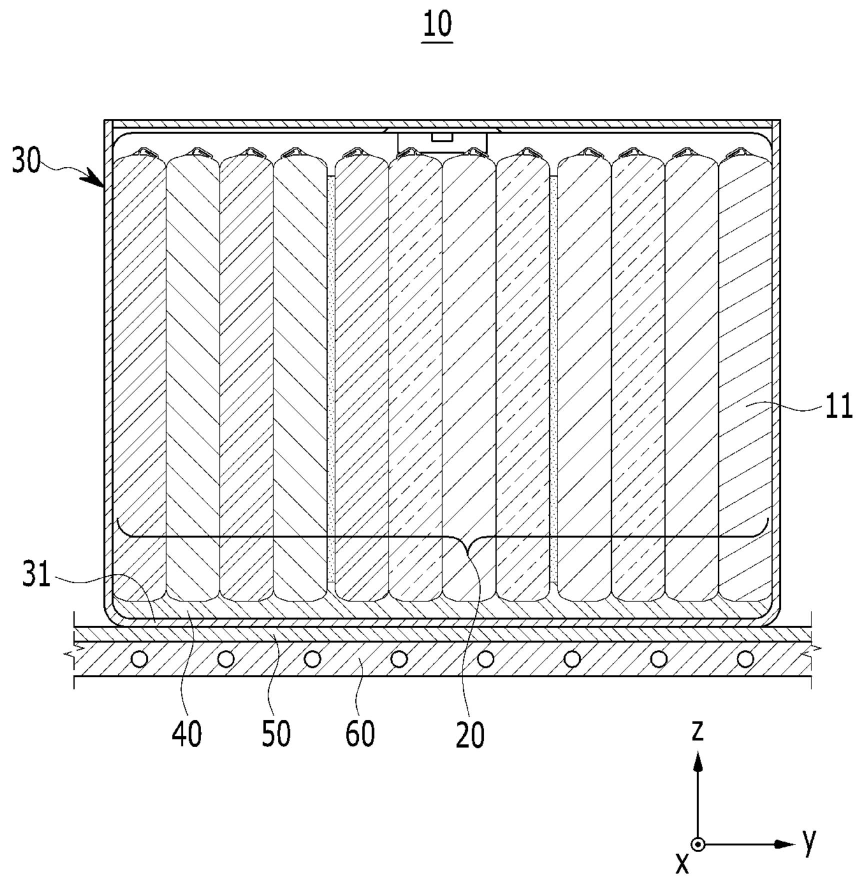


FIG. 3

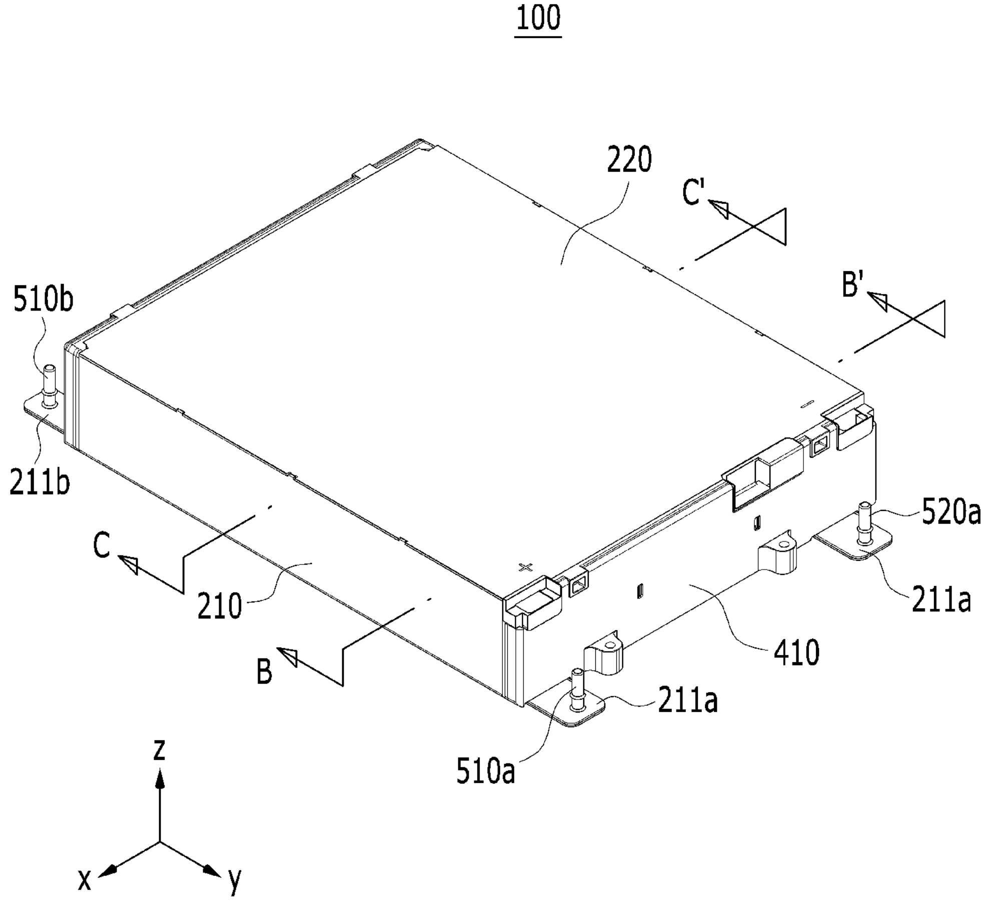


FIG. 4

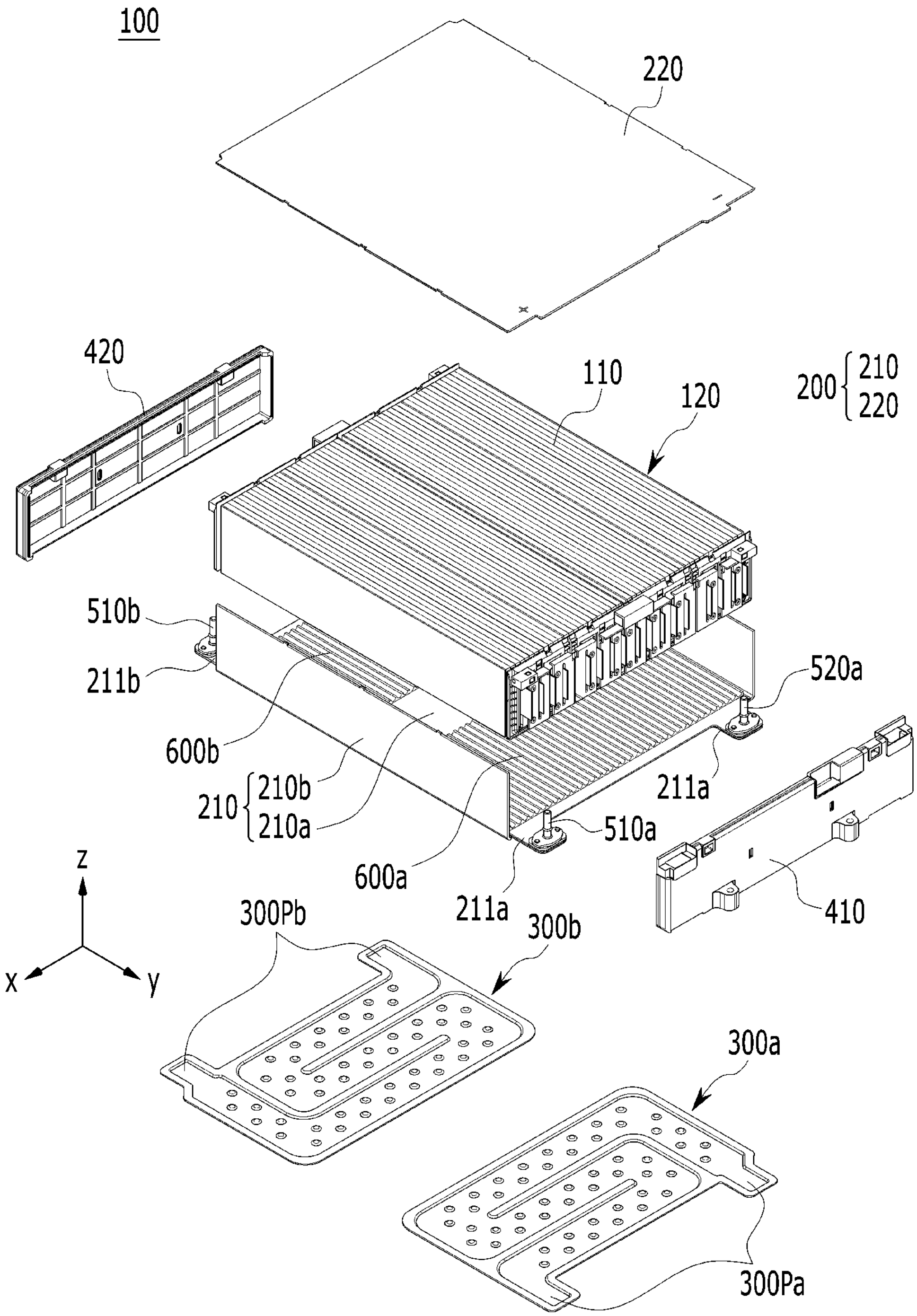

100
220
420
110
120
200
210
220
510b
211b
600b
210
210b
210a
520a
211a
510a
600a
211a
410
z
x
y
300Pb
300b
300a
300Pa

FIG. 5

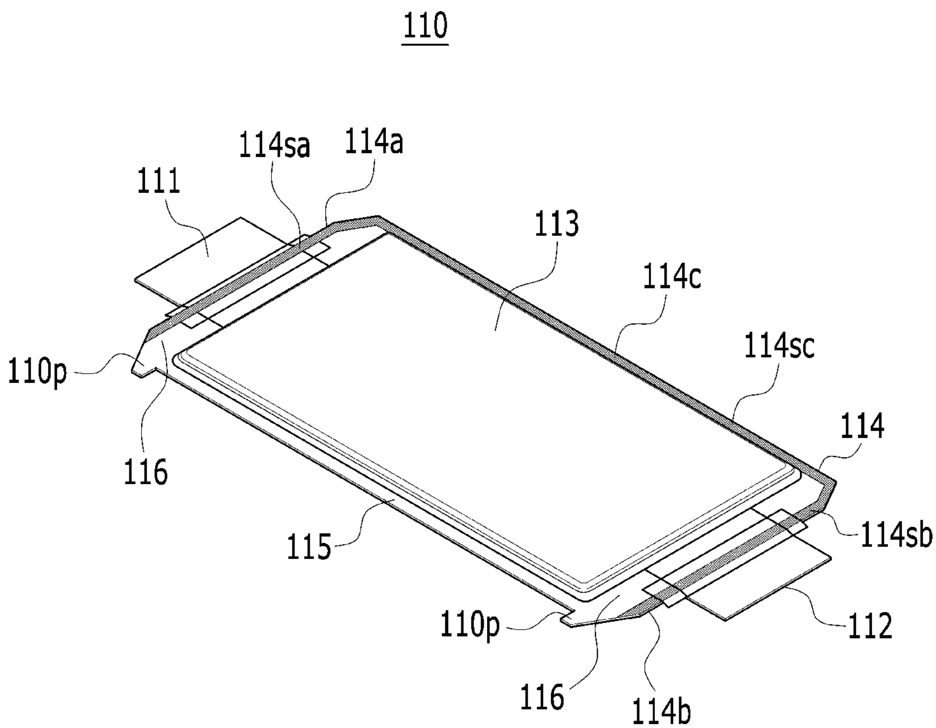

110
114sa
114a
111
113
114c
114sc
110p
114
116
114sb
115
110p
112
116
114b

FIG. 6

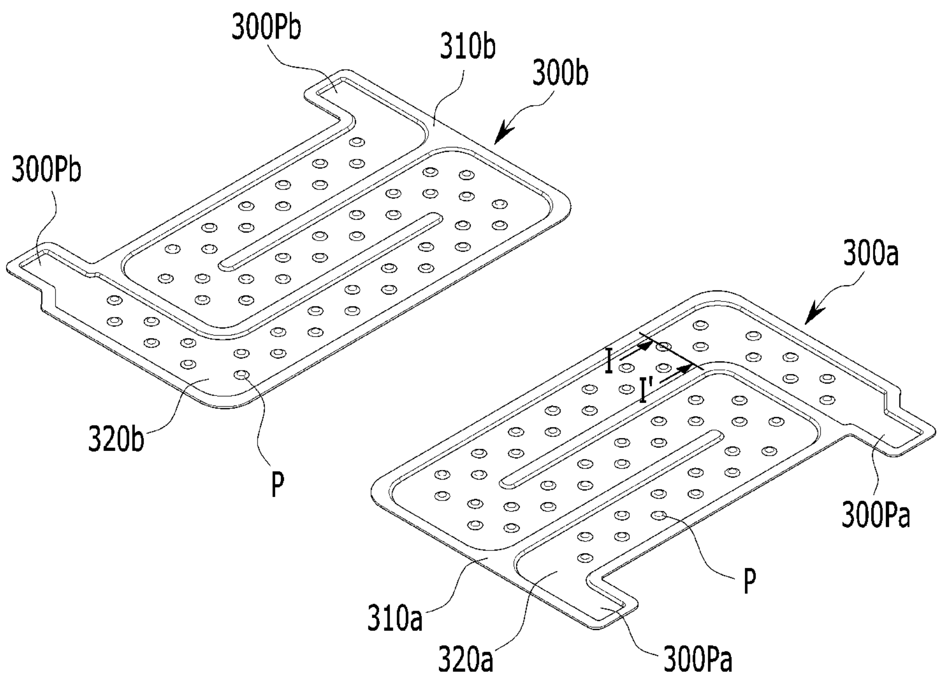


FIG. 7

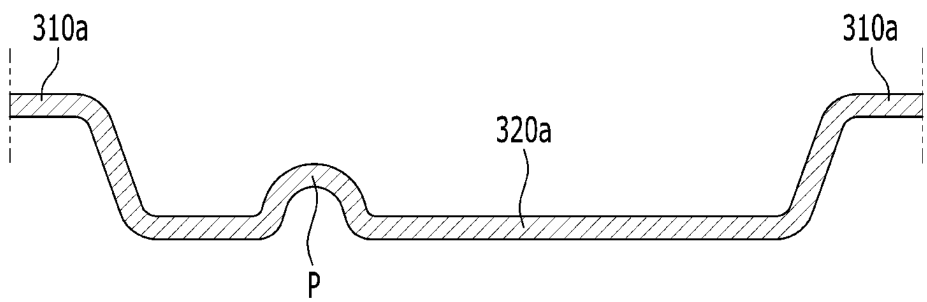
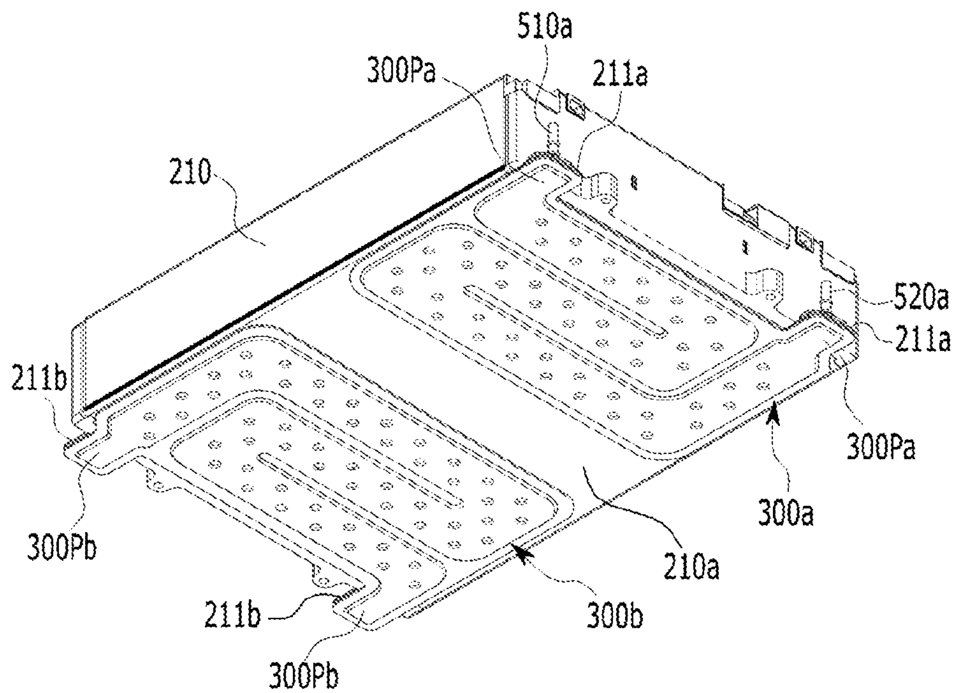


FIG. 8


510a
300Pa
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210
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211b
300Pa
300a
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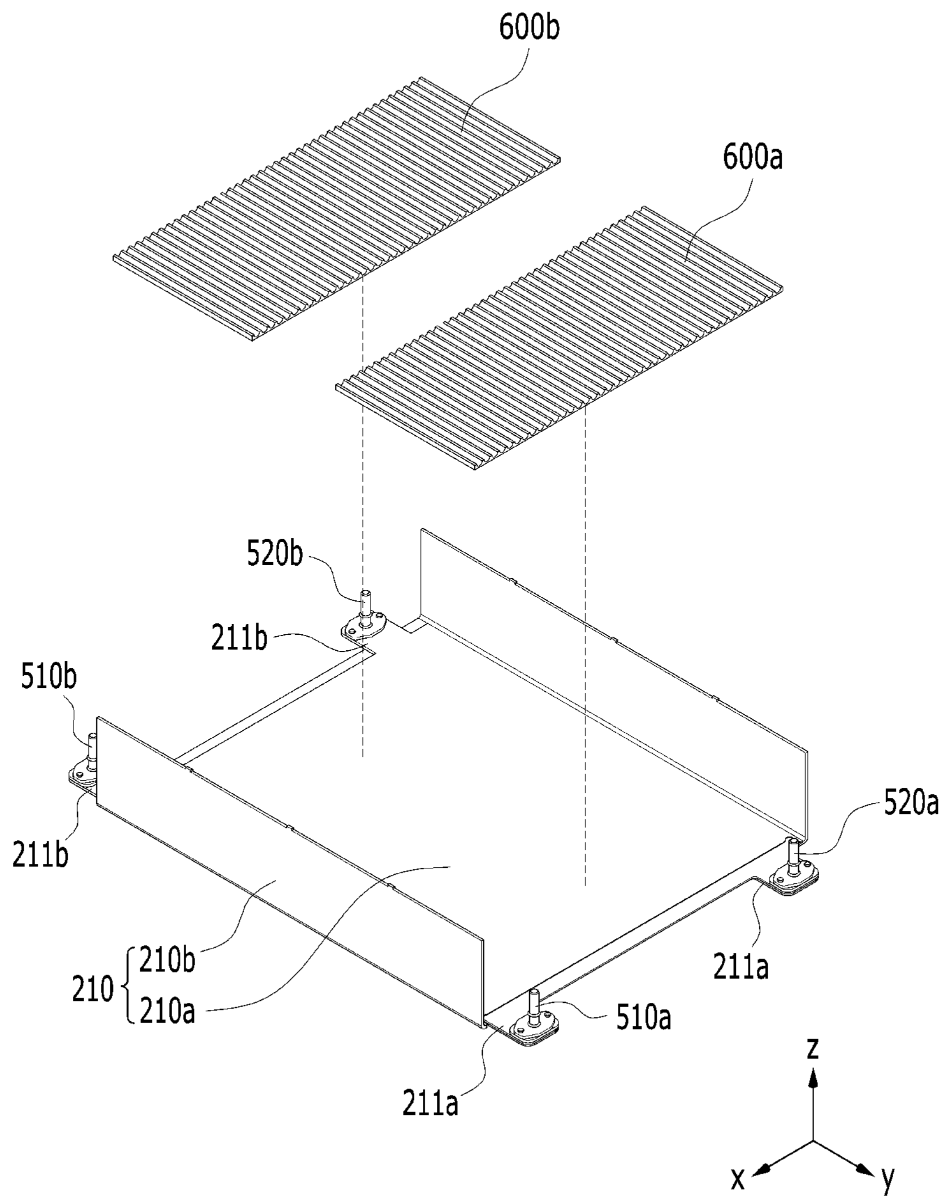
210a
211b
300b
300Pb

FIG. 9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600a
520b
211b
510b
211b
520a
210b
210
210a
211a
510a
211a
z
x
y

FIG. 10

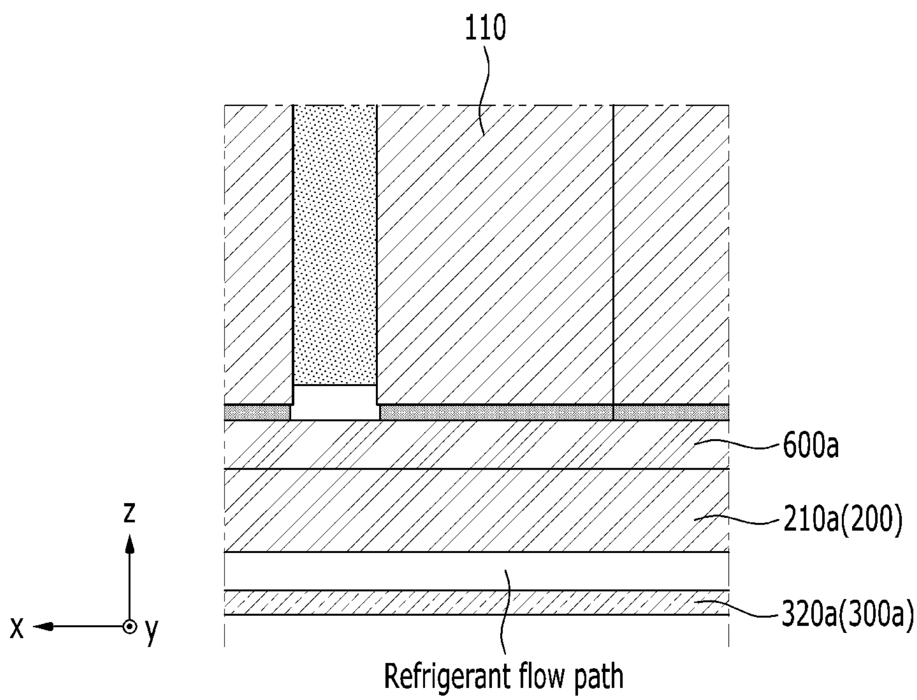


FIG. 11

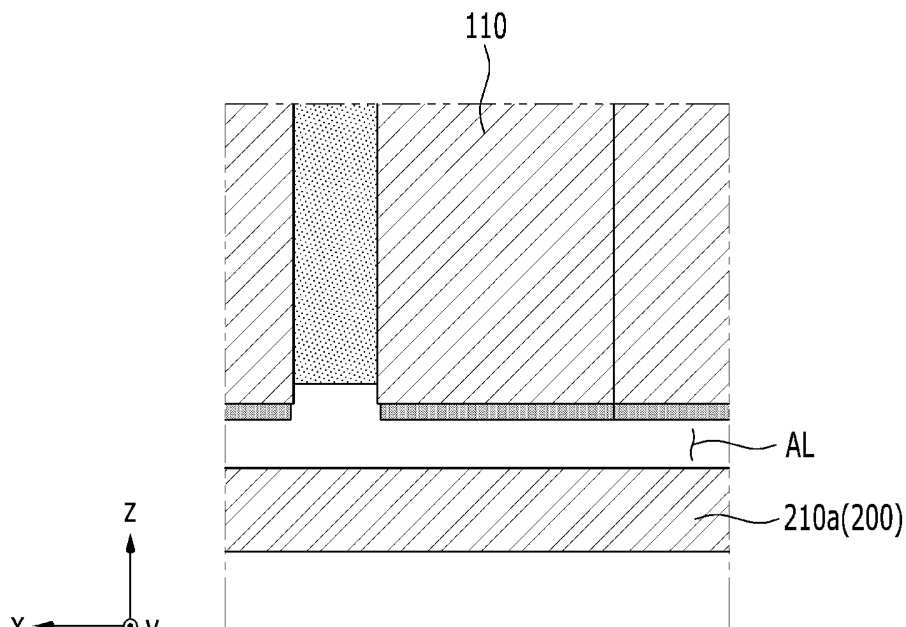


FIG. 12

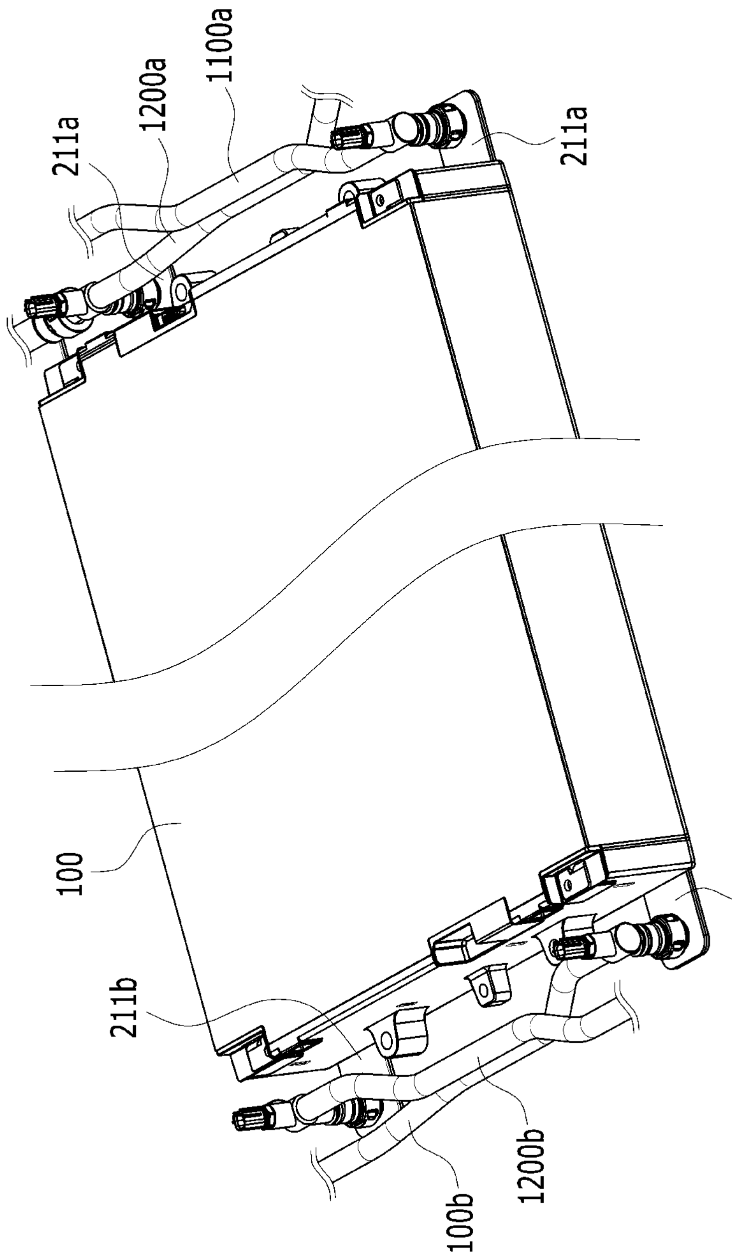


FIG. 13

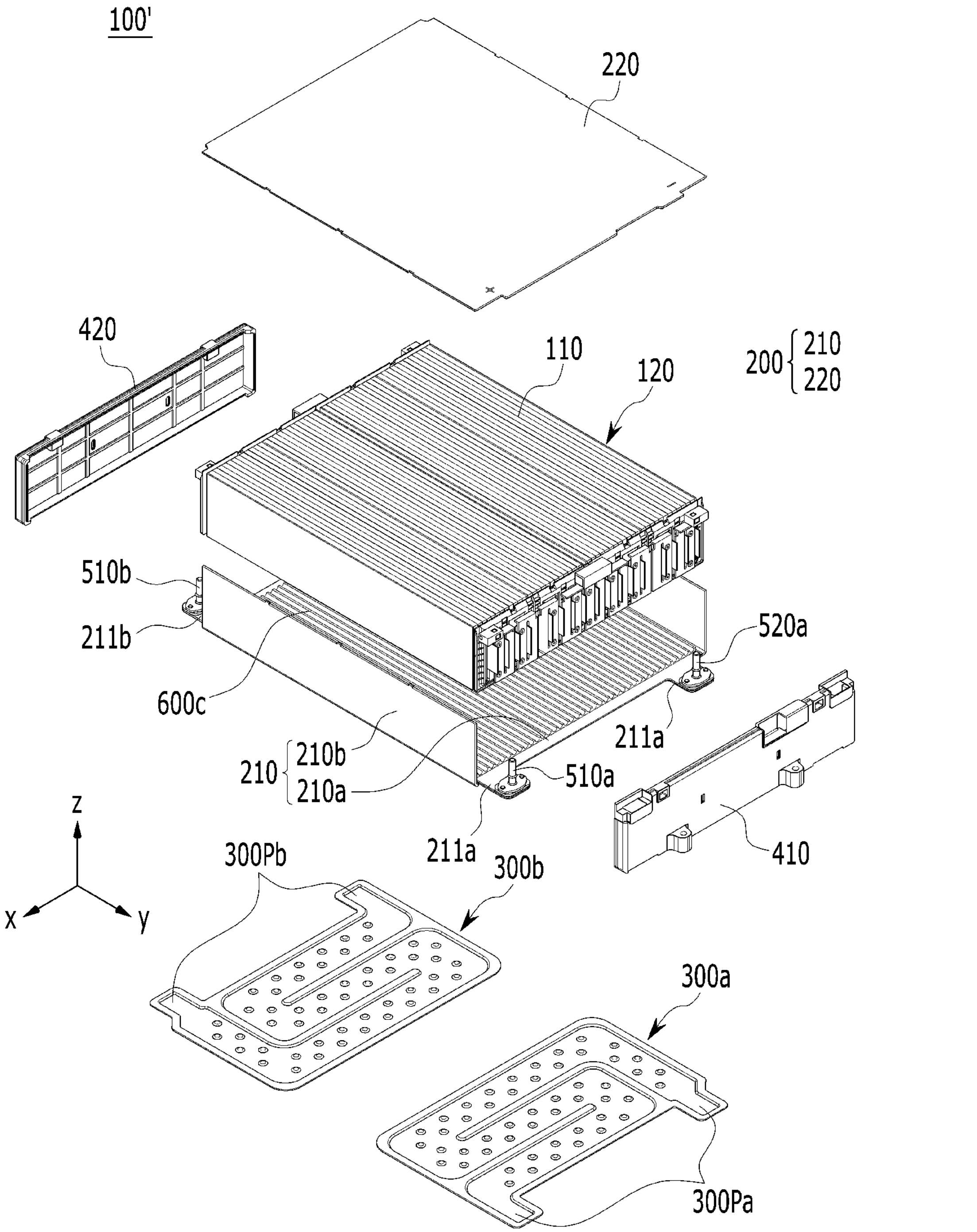
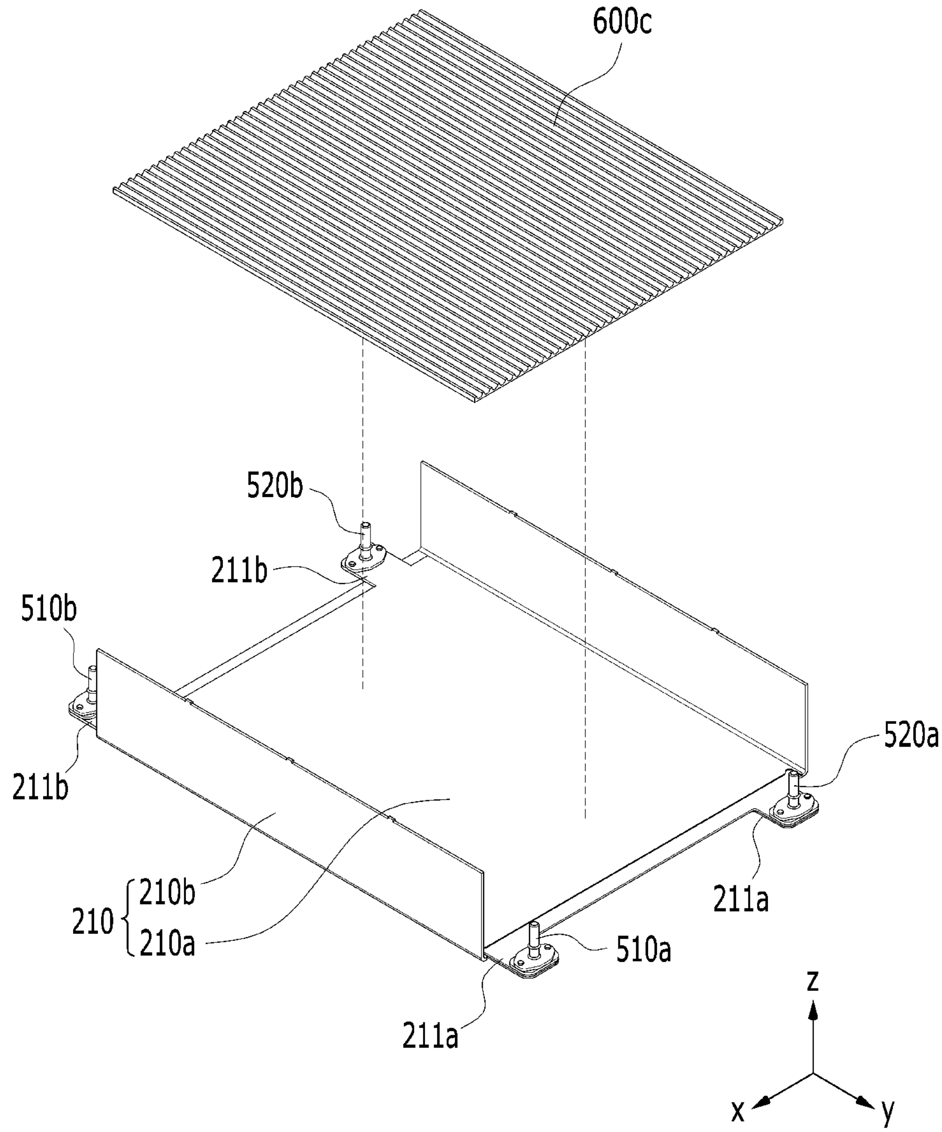

100'
220
420
110
120
200
210
220
510b
211b
600c
520a
211a
210
210b
210a
510a
211a
410
z
x
y
300Pb
300b
300a
300Pa

FIG. 14


600c
520b
211b
510b
211b
520a
210
210b
210a
211a
510a
211a
z
x
y

BATTERY MODULE AND BATTERY PACK INCLUDING THE SAME

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a US national stage of international application No. PCT/KR2022/002140 filed on Feb. 14, 2022, and claims benefit of priority to Korean patent application No. 10-2021-0024303 filed on Feb. 23, 2021, and Korean patent application No. 10-2022-0013634 filed on Jan. 28, 2022, the entire contents of which are incorporated as if fully set forth herein.

TECHNICAL FIELD

[0002] The present disclosure relates to a battery module and a battery pack including the same, and more particularly, to a battery module having improved cooling performance and a battery pack including the same.

BACKGROUND

[0003] In modern society, as portable devices such as a mobile phone, a notebook computer, a camcorder, and a digital camera are being used daily, the development of technologies in the fields related to mobile devices as described above has been activated. In addition, chargeable/dischargeable secondary batteries are used as a power source for an electric vehicle (EV), a hybrid electric vehicle (HEV), a plug-in hybrid electric vehicle (P-HEV) and the like, in an attempt to solve air pollution and the like caused by existing gasoline vehicles using fossil fuel. Therefore, there is a growing need for development of the secondary battery.

[0004] Currently commercialized secondary batteries include a nickel cadmium battery, a nickel hydrogen battery, a nickel zinc battery, and a lithium secondary battery. Among these, the lithium secondary battery has come into the spotlight because they have advantages, for example, hardly exhibiting memory effects compared to nickel-based secondary batteries and thus being freely charged and discharged and having very low self-discharge rate and high energy density.

[0005] Such lithium secondary batteries mainly use a lithium-based oxide and a carbonaceous material as a positive electrode active material and a negative electrode active material, respectively. The lithium secondary battery includes an electrode assembly in which a positive electrode plate and a negative electrode plate, each being coated with the positive electrode active material and the negative electrode active material, respectively, are disposed with a separator being interposed between them, and a battery case which seals and houses the electrode assembly together with an electrolyte solution.

[0006] Generally, the lithium secondary battery may be classified based on the shape of the exterior material into a can type secondary battery in which the electrode assembly is built into a metal can, and a pouch-type secondary battery in which the electrode assembly is built into a pouch of an aluminum laminate sheet.

[0007] In the case of a secondary battery used for small-sized devices, two to three battery cells are disposed in the secondary battery, but in the case of a secondary battery used for a medium- or large-sized device such as an automobile, a battery module in which a large number of battery cells are electrically connected is used. In such a battery module, a large number of battery cells are connected to each other in series or parallel to form a cell assembly, thereby improving capacity and output. In addition, one or more battery modules can be mounted together with various control and protection systems such as a battery management system (BMS) and a cooling system to form a battery pack.

[0008] When the temperature of the secondary battery rises higher than an appropriate temperature, the performance of the secondary battery may deteriorate, and in the worst case, there is also a risk of an explosion or ignition. In particular, heat generated from the large number of battery cells in a narrow space of a battery module or battery pack having a large number of secondary batteries, so that the temperature can rise more quickly and excessively. In other words, a battery module in which a large number of battery cells are stacked, and a battery pack equipped with such a battery module can obtain high output, but it is not easy to remove heat generated from the battery cells during charging and discharging. When the heat dissipation of the battery cells is not properly performed, deterioration of the battery cells is accelerated, the lifespan is shortened, and the possibility of explosion or ignition increases.

[0009] Moreover, a medium- or large-sized battery module included in a vehicle battery pack is frequently exposed to direct sunlight and can be subjected to high-temperature conditions such as in the summer or in desert areas.

[0010] Therefore, it may be very important to stably and effectively ensure the cooling performance when a battery module or a battery pack is configured.

[0011] FIG. **1** is a perspective view showing a conventional battery module. FIG. **2** is a cross-sectional view along the cutting line A-A' of FIG. **1**. In particular, a heat transfer member and a heat sink located under the battery module are further shown in FIG. **2**.

[0012] Referring to FIGS. **1** and **2**, the conventional battery module **10** is configured such that a plurality of battery cells **11** are stacked to form a battery cell stack **20**, and the battery cell stack **20** is housed in the housing **30**.

[0013] As described above, since the battery module **10** includes a plurality of battery cells **11**, it generates a large amount of heat in a charge and discharge process. As a cooling means, the battery module **10** may include a thermal conductive resin layer **40** that is located between the battery cell stack **20** and the bottom part **31** of the housing **30**. In addition, when the battery module **10** is mounted on a pack frame to form a battery pack, a heat transfer member **50** and a heat sink **60** may be sequentially located under the battery module **10**. The heat transfer member **50** may be a heat dissipation pad. The heat sink **60** may have a refrigerant flow passage formed therein.

[0014] The heat generated from each of the plurality of the battery cells **11** is transmitted to the outside through the thermal conductive resin layer **40**, the bottom portion **31** of the housing **30**, the heat transfer member **50**, and the heat sink **60** in this order.

[0015] In the conventional battery module **10**, the heat transfer path is complicated as described above and thus, it is difficult to effectively transfer the heat generated from the battery cells **11**. The housing **30** itself may deteriorate heat transfer properties, and a fine air layer such as an air gap, which can be formed in the space between the housing **30**, the heat transfer member **50**, and the heat sink **60**. may also be a factor that deteriorates the heat transfer properties.

[0016] As for the battery module, it can be said that it is practically necessary to develop a battery module capable of satisfying these various requirements while improving the cooling performance because other demands such as an increase in capacity are also continuing.

SUMMARY

[0017] It is an objective of the present disclosure to provide a battery module having improved cooling performance and a battery pack including the same.

[0018] However, the problem to be solved by embodiments of the present disclosure is not limited to the above-described problems, and can be variously expanded within the scope of the technical idea included in the present disclosure.

[0019] According to one embodiment of the present disclosure, there is provided a battery module comprising: a battery cell stack in which a plurality of battery cells including electrode leads protruding in mutually opposite directions are stacked; a housing that houses the battery cell stack; a first heat sink and a second heat sink that are located under the bottom part of the housing; first and second refrigerant flow paths formed between the first heat sink and the bottom part of the housing and between the second heat sink and the bottom part of the housing, respectively, wherein the first refrigerant flow path formed by the first heat sink is separated from the second refrigerant flow path formed by the second heat sink.

[0020] The first heat sink and the second heat sink may be located separately from each other along a direction parallel to a direction in which the electrode leads protrude.

[0021] The battery module may further comprise a first thermal conductive resin layer and a second thermal conductive resin layer that are located between the lower surface of the battery cell stack and the bottom part of the housing.

[0022] The first thermal conductive resin layer and the second thermal conductive resin layer may be located separately from each other along a direction parallel to a direction in which the electrode leads protrude.

[0023] The first heat sink may be located at a portion corresponding to the first thermal conductive resin layer, and the second heat sink is located at a portion corresponding to the second thermal conductive resin layer, with respect to a direction perpendicular to one surface of the bottom part of the housing.

[0024] The first heat sink may comprise a first lower plate that is joined to the bottom part of the housing, and a first recessed part that is recessed in a lower direction from the first lower plate.

[0025] The battery module may further comprise a thermal conductive resin layer located between the lower surface of the battery cell stack and the bottom part of the housing.

[0026] The thermal conductive resin layer may cover a region where the first heat sink is located and a region where the second heat sink is located, with respect to a direction perpendicular to one surface of the bottom part of the housing.

[0027] The housing may comprise one or more first housing protrusion parts that protrude from a part of the bottom part of the housing, and a refrigerant injection port that may be connected to any one of the first housing protrusion parts and a refrigerant discharge port that may be connected to any other one of the protrusion parts.

[0028] The first heat sink may comprise first heat sink protrusion parts that protrudes from one side of the first heat sink to portions where the first housing protrusion parts are located.

[0029] The second heat sink may comprise a second lower plate that is joined to the bottom part of the housing, and a second recessed part that is recessed in a lower direction from the second lower plate.

[0030] The housing may comprise one or more second housing protrusion parts that protrude from a part of the bottom part of the housing, and a refrigerant injection port that may be connected to any one of the second housing protrusion parts, and a refrigerant discharge port that may be connected to any other one of the second housing protrusion parts.

[0031] The second heat sink may comprise second heat sink protrusion parts that protrudes from one side of the second heat sink to portions where the second housing protrusion parts are located.

[0032] According to another embodiment of the present disclosure, there is provided a battery pack comprising: the battery module; a first pack refrigerant supply tube that supplies a refrigerant to a refrigerant flow path formed by a first heat sink; a first pack refrigerant discharge tube that discharges the refrigerant from the refrigerant flow path formed by the first heat sink; a second pack refrigerant supply tube that supplies a refrigerant to the refrigerant flow path formed by the second heat sink; and a second pack refrigerant discharge tube that discharges the refrigerant from the refrigerant flow path formed by the second heat sink.

[0033] According to embodiments of the present disclosure, the thermal conductive resin layer and the heat sink are concentrically arranged in the portion where heat generation is excessive in the battery cells, thereby being capable of improving the cooling performance and reducing the temperature deviation of the battery cells.

[0034] Additionally, the length of the refrigerant flow path of the heat sink relative to the cooling area of the battery module can be shortened, and thus the pressure drop of the refrigerant flow path can be improved.

[0035] Moreover, the cooling performance can be improved through the integrated structure of the housing and the heat sink.

[0036] Further, by removing unnecessary cooling structures, it is possible to reduce costs, improve space utilization, and increase the capacity or output of the battery module.

[0037] The effects of the present disclosure are not limited to the effects mentioned above and additional other effects not described above will be clearly understood from the description of the appended claims by those skilled in the art.

BRIEF DESCRIPTION OF THE DRAWINGS

[0038] FIG. **1** is a perspective view of a conventional battery module;

[0039] FIG. **2** is a cross-sectional view along line A-A' of FIG. **1**;

[0040] FIG. **3** is a perspective view of a battery module according to an embodiment of the present disclosure;

[0041] FIG. **4** is an exploded perspective view of the battery module of FIG. **3**;

[0042] FIG. **5** is a perspective view of one of the battery cells included in the battery module of FIG. **4**;

[0043] FIG. **6** is a perspective view of a first heat sink and a second heat sink included in the battery module of FIG. **4**;

[0044] FIG. **7** is a cross-sectional view along line I-I' of FIG. **6**;

[0045] FIG. **8** is a perspective view of the battery module as viewed from the bottom to the top showing the lower surface of the battery module of FIG. **3**;

[0046] FIG. **9** is a perspective of a lower frame, a first thermal conductive resin layer, and a second thermal conductive resin layer included in the battery module of FIG. **4**;

[0047] FIG. **10** is a cross-sectional view along line B-B' of FIG. **3**;

[0048] FIG. **11** is a cross-sectional view along line C-C' of FIG. **3**;

[0049] FIG. **12** is a perspective view of a battery module, a pack refrigerant supply tube, and a pack refrigerant discharge tube according to an embodiment of the present disclosure;

[0050] FIG. **13** is an exploded perspective view of a battery module according to another embodiment of the present disclosure; and

[0051] FIG. **14** is a perspective view of a lower frame and a thermal conductive resin layer included in the battery module of FIG. **13**.

DETAILED DESCRIPTION

[0052] Hereinafter, various embodiments of the present disclosure will be described in detail with reference to the accompanying drawings so that those skilled in the art can easily carry out these embodiments. The present disclosure may be modified in various different ways and is not limited to the embodiments set forth herein.

[0053] A description of parts not related to the description will be omitted herein for clarity, and like reference numerals designate like elements throughout the description.

[0054] Further, in the drawings, the size and thickness of each element are arbitrarily illustrated for convenience of description, and the present disclosure is not necessarily limited to those illustrated in the drawings. In the drawings, the thickness of layers, regions, etc. are exaggerated for clarity. In the drawings, for convenience of description, the thicknesses of some layers and regions are exaggerated.

[0055] In addition, it will be understood that when an element such as a layer, film, region, or plate is referred to as being "on" or "above" another element, it can be directly on the other element or intervening elements may also be present. In contrast, when an element is referred to as being "directly on" another element, it means that other intervening elements are not present. Further, the word "on" or "above" means disposed on or below a reference portion, and does not necessarily mean being disposed on the upper end of the reference portion toward the opposite direction of gravity.

[0056] Further, throughout the description, when a portion is referred to as "including" a certain component, it means that the portion can further include other components, without excluding the other components, unless otherwise stated.

[0057] Further, throughout the description, when referred to as "planar", it means when a target portion is viewed from the upper side, and when referred to as "cross-sectional", it means when a target portion is viewed from the side of a cross section cut vertically.

[0058] FIG. **3** is a perspective view of a battery module according to an embodiment of the present disclosure. FIG. **4** is an exploded perspective view of the battery module of FIG. **3**. FIG. **5** is a perspective view of one of the battery cells included in the battery module of FIG. **4**.

[0059] Referring to FIGS. **3** to **5**, a battery module **100** includes a battery cell stack **120** in which a plurality of battery cells **110** are stacked; a housing **200** that houses the battery cell stack **120**; and a first heat sink **300*a*** and a second heat sink **300*b*** that are located under the bottom part **210*a*** of the housing.

[0060] First, the battery cell **110** is preferably a pouch type battery cell and may be formed in a rectangular sheet-like structure. For example, the battery cell **110** according to the present embodiment has a structure in which two electrode leads **111** and **112** face each other and protrude from one end **114*a*** and the other end **114*b*** of a cell body **113**, respectively. That is, the battery cell **110** includes electrode leads **111** and **112** protruding in mutually opposite directions. More specifically, the first and second electrode leads **111** and **112** are connected to an electrode assembly (not shown) and protrude from the electrode assembly (not shown) to the outside of the battery cell **110**.

[0061] Meanwhile, the battery cell **110** can be produced by joining both end parts **114*a*** and **114*b*** of a cell case **114** and one side part **114*c*** connecting them in a state in which an electrode assembly (not shown) is housed in a cell case **114**. In other words, the battery cell **110** according to the present embodiment has a total of three sealing parts **114*sa***, **114*sb*** and **114*sc***, the sealing parts **114*sa***, **114*sb*** and **114*sc*** have a structure sealed by a method such as heat fusion, and the remaining other side part may be composed of a connection part **115**. The cell case **114** may be composed of a laminate sheet including a resin layer and a metal layer.

[0062] In addition, the connection part **115** may extend long along one edge of the battery cell **110**, and a bat-ear **110*p*** may be formed at an end part of the connection part **115**. Further, while the cell case **114** is sealed with the protruding electrode leads **111** and **112** being interposed at ends thereof, a terrace part **116** may be formed between the electrode leads **111** and **112** and the cell body **113**. That is, the battery cell **110** includes a terrace part **116** formed to extend from the cell case **114** in a protruding direction of the electrode leads **111** and **112**.

[0063] A plurality of battery cells **110** may be present, and the plurality of battery cells **110** may be stacked to be electrically connected to each other, thereby forming a battery cell stack **120**. Particularly, as shown in FIG. **4**, a plurality of battery cells **110** may be stacked along a direction parallel to the x-axis. Thereby, the electrode leads **111** and **112** may protrude in the +y-axis direction and the −y-axis direction, respectively. Although not specifically shown in the figure, an adhesive member may be located between the battery cells **110**. Thereby, the battery cells **110** may be adhered to each other to form the battery cell stack **120**.

[0064] The battery cell stack **120** according to the present embodiment may be a large-area module in which the number of battery cells **110** is larger than that of the conventional case. Specifically, 32 to 48 battery cells **110** can be included per battery module **100**. In the case of such a large-area module, the horizontal length of the battery module becomes long. Here, the horizontal length may mean a length in the direction in which the battery cells **110** are stacked, that is, in a direction parallel to the x-axis.

[0065] When the battery cell **110** is repeatedly charged and discharged, heat is generated, and a large amount of heat is generated in the portions adjacent to the electrode leads **111** and **112**. That is, more heat is generated during charge and discharge as approaching the terrace part **116** rather than the central part of the cell body **113**.

[0066] The housing **200** for housing the battery cell stack **120** may include a lower frame **210** and an upper cover **220**.

[0067] The lower frame **210** may include a bottom part **210***a* and two side parts **210***b* extending upward at both ends of the bottom part **210***a*. The bottom part **210***a* may cover the lower surface of the battery cell stack **120**, and the two side parts **210***b* may cover both side surfaces of the battery cell stack **120**. Here, the lower surface of the battery cell stack **120** refers to a surface in the −z-axis direction, and both side surfaces of the battery cell stack **120** refer to surfaces in the +x-axis and −x-axis directions. However, these are surfaces mentioned for convenience of explanation, and may differ depending on the position of a target object or the position of an observer, and the like.

[0068] The upper cover **220** may have a single plate-shaped structure that wraps the remaining upper surface (surface in the z-axis direction) excluding the lower surface and both side surfaces of the battery cell stack **120** that is wrapped by the lower frame **210**. The upper cover **220** and the lower frame **210** are joined by welding in the state where mutually corresponding edge parts are in contact with each other, thereby forming a structure wrapping the battery cell stack **120** vertically and horizontally. The battery cell stack **120** can be physically protected through the upper cover **220** and the lower frame **210**. For this purpose, the upper cover **220** and the lower frame **210** may include a metal material to have a predetermined strength.

[0069] Meanwhile, although not specifically shown in the figure, the housing **100** according to a modified embodiment of the present disclosure may be a mono frame in the form of a metal plate in which the upper part, the lower part, and both side parts are integrated. That is, this frame may have a structure in which the upper part, the lower part, and both side parts are integrated by being manufactured using extrusion molding, rather than a structure in which the lower frame **210** and the upper cover **220** are joined with each other.

[0070] Meanwhile, the battery module **100** according to the present embodiment may include a first end plate **410** and a second end plate **420** each covering the front surface and the rear surface of the battery cell stack **120**, respectively. Here, the front surface of the battery cell stack **120** refers to a surface in the +y-axis direction, and the rear surface of the battery cell stack **120** refers to a surface in the −y-axis direction.

[0071] The first end plate **410** and the second end plate **420** are located at both open sides of the housing **200** so that they can be formed to cover the battery cell stack **120**. Each of the first end plate **410** and the second end plate **420** may be joined by welding in the configuration where corresponding edges of the housing **200** are in contact with each other. The first end plate **410** and the second end plate **420** can physically protect the battery cell stack **120** and other electrical components from external impacts.

[0072] Meanwhile, although not specifically shown in the figure, a bus bar frame to which the bus bar is mounted and an insulating cover for electrical insulation can be located between the battery cell stack **120** and each of the end plates **410** and **420**.

[0073] Next, the first and second heat sinks according to the present embodiment will be described in detail with reference to FIGS. **4** and **6** to **9**.

[0074] FIG. **6** is a perspective view of a first heat sink and a second heat sink included in the battery module of FIG. **4**. FIG. **7** is a cross-sectional view along the line I-I' of FIG. **6**. FIG. **8** is a perspective view of the battery module when viewed from the bottom to the top so that the lower surface of the battery module of FIG. **3** can be seen. FIG. **9** is a perspective view of a lower frame, a first thermal conductive resin layer, and a second thermal conductive resin layer included in the battery module of FIG. **4**.

[0075] Referring to FIG. **4** and FIGS. **6** to **9**, the first heat sink **300***a* and the second heat sink **300***b* according to the present embodiment are located under the bottom part **210***a* of the housing **200**, and first and second refrigerant flow paths are formed between the first heat sink **300***a* and the bottom part **210***a* of the housing **200** and between the second heat sink **300***b* and the bottom part **210***a* of the housing **200**, respectively. The first refrigerant flow path formed by the first heat sink **300***a* and the second refrigerant flow path formed by the second heat sink **300***b* are separated from each other. In other words, the first refrigerant flowing inside the first heat sink **300***a* and the second refrigerant flowing inside the second heat sink **300***b* flow along different paths from each other without mixing. Here, the refrigerant is a medium for cooling, and is not particularly limited, but may be a cooling water.

[0076] The first heat sink **300***a* and the second heat sink **300***b* may be located separately from each other in a direction parallel to the direction in which the electrode leads **111** and **112** protrude. As described above, the electrode leads **111** and **112** in the battery cell stack **120** can protrude in the +y-axis direction and the −y-axis direction, respectively. The first heat sink **300***a* and the second heat sink **300***b* may be located separately from each other along a direction parallel to the +y-axis to comply with the protruding directions of the electrode leads **111** and **112**. In particular, the first heat sink **300***a* and the second heat sink **300***b* may be located to be adjacent to both sides facing each other at the bottom part **210***a* of the housing **200**.

[0077] Referring to FIG. **5**, as described above, a portion closer to the terrace part **116** than the central part of the cell body **113** of in the battery cell **110** generates more heat during charge and discharge. With reference to the battery cell stack **120**, a large amount of heat is generated in a portion adjacent to the portion where the electrode leads **111** and **112** protrude. As the charge and discharge cycles are repeated, the temperature deviation between the respective portions of the battery cell **110** becomes larger. It is important to eliminate the temperature deviation because this temperature deviation leads to deterioration of battery performance.

[0078] Referring to FIGS. **4** and **6** together with FIG. **5**, the battery module **100** according to the present embodiment attempts to concentrate the cooling function and eliminate the temperature deviation of the battery cell **110** by providing the first heat sink **300***a* and the second heat sink **300***b* in a portion where heat generation is excessive in the battery cell **110**. Thereby, in the battery module **100** according to the present embodiment, heat dissipation can be effectively made at both end parts of the battery cell **110** where heat generation is excessive, and the temperature deviation between the respective portions of the battery cell **110** can be minimized.

[0079] Next, specific structures of the first heat sink **300***a* and the second heat sink **300***b* will be described.

[0080] The first heat sink **300***a* includes a first lower plate **310***a* that is joined to the bottom part **210***a* of the housing **200**, and a first recessed part **320***a* that is recessed in a lower direction from the second lower plate **310***a*. The first lower plate **310***a* forms a skeleton of the first heat sink **300***a* and can be directly joined to the bottom part **210***a* of the housing **200** by a welding method. Specifically, as shown in FIG. **7**, the first recessed part **320***a* is a part that is recessed in a lower direction, and a path through which the refrigerant flows may be formed in the inside thereof.

[0081] The first recessed part **320***a* may be a tube in which a cross section cut in the xz plane or the yz plane perpendicular to the direction in which the refrigerant flow path extends is U-shaped, and the bottom part **210***a* may be located on the open upper side of the U-shaped tube. That is, as shown in FIG. **7**, the first recessed part **320***a* may be a tube having a U-shaped cross-section. A second recessed part **320***b* described later also has a similar structure. As the first lower plate **310***a* comes into contact with the bottom part **210***a*, the space between the first recessed part **320***a* and the bottom part **210***a* becomes a region through which the refrigerant flows, that is, a flow path of the refrigerant. Thereby, the bottom part **210***a* of the housing **200** may be in direct contact with the refrigerant.

[0082] Meanwhile, the housing **200** may include first housing protrusion parts **211***a* formed in a part of the bottom part **210***a* of the housing **200**. Specifically, the first housing protrusion parts **211***a* may be formed such that a part of the bottom part **210***a* of the housing **200** extends and passes through the first end plate **410**. In particular, at least two first housing protrusion parts **211***a* may be formed, and one of the two first housing protrusion parts **211***a* may be connected to a refrigerant injection port **510***a*, and another one of the two first housing protrusion parts **211***a* may be connected to a refrigerant discharge port **520***a*. The refrigerant injection port **510***a* and the refrigerant discharge port **520***a* may be connected to a first pack refrigerant supply tube and a first pack refrigerant discharge tube, respectively, each of which will be described later.

[0083] The first heat sink **300***a* may include first heat sink protrusion parts **300**Pa that protrudes from one side of the first heat sink **300***a* to a portion where the first housing protrusion parts **211***a* are located. At least two first housing protrusion parts **211***a* may be formed, and the first heat sink protrusion parts **300**Pa may be formed to correspond to each of the first housing protrusion parts **211***a*.

[0084] The first recessed part **320***a* may be connected from one of the first heat sink protrusions **300**Pa to another one of the first heat sink protrusions **300**Pa. The refrigerant supplied via the refrigerant injection port **510***a* passes between the first housing protrusion part **211***a* and the first heat sink protrusion part **300**Pa, and first flows into the space between the first recessed part **320***a* and the bottom part **210***a*. After that, the refrigerant moves along the first recessed part **320***a* and passes through the other first housing projection part **211***a* and the other first heat sink projection part **300**Pa to be discharged via the refrigerant discharge port **520***a*.

[0085] Meanwhile, the second heat sink **300***b* includes a second lower plate **310***b* that is joined to the bottom part **210***a* of the housing **200**, and a second recessed part **320***b* that is recessed in a lower direction from the second lower plate **310***b*. The second lower plate **310***b* forms a skeleton of the second heat sink **300***b* and may be directly joined to the bottom part **210***a* of the housing **200** by a welding method. The second recessed part **320***b* is a part recessed in the lower direction, and a path through which the refrigerant flows may be formed in the inside thereof.

[0086] The second recessed part **320***b* may be a tube having a U-shaped vertical cross-section in the xz plane or the yz plane with respect to the direction in which the refrigerant flow path extends, and the bottom part **210***a* may be located on the open upper side of the U-shaped tube. As the second lower plate **310***b* comes into contact with the bottom part **210***a*, the space between the second recessed part **320***b* and the bottom part **210***a* becomes a region through which the refrigerant flows, that is, a flow path of the refrigerant. Thereby, the bottom part **210***a* of the housing **200** may be in direct contact with the refrigerant.

[0087] Meanwhile, the housing **200** may include second housing protrusion parts **211***b* formed in a part of the bottom part **210***a* of the housing **200**. Specifically, the second housing protrusion parts **211***b* may be formed such that a part of the bottom part **210***a* of the housing **200** extends and passes through the second end plate **420**. In particular, at least two second housing protrusion parts **211***b* may be formed, and one of the two second housing protrusion parts **211***b* may be connected to a refrigerant injection port **510***b*, and the other of the two second housing protrusion parts **211***b* may be connected to a refrigerant discharge port **520***b*. The refrigerant injection port **510***b* and the refrigerant discharge port **520***b* may be connected to a second pack refrigerant supply tube and a second pack refrigerant discharge tube, respectively, each of which will be described later. At this time, the second housing protrusion parts **211***b* may be formed on sides of the bottom part **210***a* on which the first housing protrusion parts **211***a* are not formed. More specifically, the second housing protrusion parts **211***b* may be formed on a side opposite to the side of the bottom part **210***a* on which the first housing protrusion parts **211***a* are formed.

[0088] The second heat sink **300***b* may include second heat sink protrusions parts **300**Pb that protrude from one side of the second heat sink **300***b* to a portion where the second housing protrusion parts **211***b* are located. At least two second housing protrusion parts **211***b* can be formed, and the second heat sink protrusion parts **300**Pb may be formed to correspond to each of the second housing protrusion parts **211***b*.

[0089] The second recessed part **320***b* may be connected from one of the second heat sink protrusion parts **300**Pb to the other one of the second heat sink protrusion parts **300**Pb. The refrigerant supplied via the refrigerant injection port **510***b* passes between the second housing protrusion part **211***b* and the second heat sink protrusion part **300**Pb, and first flows into the space between the second recessed part **320***b* and the bottom part **210***a*. After that, the refrigerant moves along the second recessed part **320***b* and passes through the other second housing protrusion **211***b* and the second heat sink protrusion **300**Pb to be discharged via the refrigerant discharge port **520***b*.

[0090] In this manner, the first refrigerant flow path formed by the first recessed part **320***a* of the first heat sink **300***a*, and the second refrigerant flow path formed by the second recessed part **320***b* of the second heat sink **300***b* are separated from each other. In particular, as shown in FIG. **8**, instead of covering the entire lower surface of the bottom part **210***a* with a single heat sink, the first heat sink **300***a* and the second heat sink **300***b* separated from each other are separately covered. If only a single heat sink is disposed, the path from where the refrigerant flows in to where the refrigerant is discharged becomes longer. If the path becomes longer, there is a problem that the pressure in the latter half of the refrigerant flow path decreases. Further, the cooling performance deteriorates in the latter half of the refrigerant flow path because heat exchange via the refrigerant has already proceeded to a considerable extent.

[0091] On the other hand, in the battery module **100** according to the present embodiment, the refrigerant path of each heat sink is shortened by disposing the first and second heat sinks **300***a* and **300***b*. Therefore, the pressure drop problem in the latter half can be solved, and the cooling performance of each heat sink can be concentrated on the area where heat generation is excessive, so that more effective heat dissipation is possible. In particular, as in the present embodiment, in the case of a large-area module in which the number of battery cells **110** is larger than that of the conventional module, it is more effective in terms of pressure maintenance and cooling performance of the refrigerant flow path than when only a single heat sink is disposed.

[0092] Further, when a large-area battery module including a single heat sink is disposed in a device such as an automobile, a large-capacity refrigerant pump is required to supply and discharge refrigerant into the single heat sink. Since such a large-capacity refrigerant pump occupies a large space, the space efficiency inside a device such as an automobile deteriorates.

[0093] On the other hand, when the separated first and second heat sinks **300***a* and **300***b* are included as in the battery module according to the present embodiment, equivalent heat exchange performance and cooling performance can be realized even with a refrigerant pump with a smaller capacity. That is, since a refrigerant pump with a smaller capacity can be used, there is an advantage that the space inside a device such as an automobile can be efficiently utilized.

[0094] Meanwhile, at least one of the first recessed part **320***a* and the second recessed part **320***b* may have a protrusion pattern P protruding in an upper direction. As in the battery cell stack **120** according to the present embodiment, in the case of a large-area battery module in which the number of stacked battery cells increases significantly compared to the conventional module, the width of the refrigerant flow path may be wider and thus, the temperature deviation may be more excessive. The protrusion pattern P can generate the effect of substantially reducing the width of the cooling flow path, minimize the pressure drop, and at the same time, reduce the temperature deviation between the widths of the refrigerant flow paths. Therefore, a uniform cooling effect can be realized.

[0095] Next, the first thermal conductive resin layer and the second thermal conductive resin layer according to an embodiment of the present disclosure will be described in detail with reference to FIGS. **9** to **11**.

[0096] FIG. **10** is a cross-sectional view along line B-B' of FIG. **3**. FIG. **11** is a cross-sectional view along line C-C' of FIG. **3**.

[0097] Referring to FIGS. **3**, **4** and **9** to **11** together, the battery module **100** according to the present embodiment may further include a first thermal conductive resin layer **600***a* and a second thermal conductive resin layer **600***b* that are located between the lower surface of the battery cell stack **120** and the bottom part **210***a* of the housing **200**.

[0098] The first thermal conductive resin layer **600***a* and the second thermal conductive resin layer **600***b* may include a thermal conductive resin. The thermal conductive resin may be applied to the bottom part **210***a* of the housing **200** to form a first thermal conductive resin layer **600***a* and a second thermal conductive resin layer **600***b*. The thermal conductive resin may include a thermal conductive adhesive material, and specifically, may include at least one of a silicone material, a urethane material, and an acrylic material. The thermal conductive resin is a liquid during application but is cured after application, so that it can perform the role of fixing one or more battery cells **110** constituting the battery cell stack **120**. Further, heat generated from the battery cell **100** can be quickly transferred to the outside of the battery module **100** because the thermal conductive resin has excellent heat transfer properties, thereby preventing overheating of the battery pack.

[0099] The first thermal conductive resin layer **600***a* and the second thermal conductive resin layer **600***b* may be located separately from each other in a direction parallel to the direction in which the electrode leads **111** and **112** protrude. As described above, in the battery cell stack **120**, the electrode leads **111** and **112** may protrude in the +y-axis direction and the −y-axis direction, respectively. The first thermal conductive resin layer **600***a* and the second thermal conductive resin layer **600***b* may be located separately from each other in a direction parallel to the y-axis to comply with the protruding directions of the electrode leads **111** and **112**. In particular, the first thermal conductive resin layer **600***a* and the second thermal conductive resin layer **600***b* may be located to be adjacent to both sides facing each other at the bottom part **210***a* of the housing **200**, respectively. Further, with respect to a direction perpendicular to one surface of the bottom part **210***a* of the housing **200**, the first heat sink **300***a* may be located at a portion corresponding to the first thermal conductive resin layer **600***a*, and the second heat sink **300***b* may be located at a portion corresponding to the second thermal conductive resin layer **600***b*.

[0100] Similar to the first heat sink **300***a* and the second heat sink **300***b*, the battery module **100** according to the present embodiment provides the first thermal conductive resin layer **600***a* and the second thermal conductive resin layer **600***b* in a portion where heat generation is excessive in the battery cell **110**, thereby attempting to concentrate the cooling function and eliminate the temperature deviation of the battery cell **110**. Thereby, in the battery module **100** according to the present embodiment, heat can be effectively dissipated at both end parts of the battery cell **110** where heat generation is excessive, and the temperature deviation between the respective parts of the battery cell **110** can be minimized.

[0101] Specifically, as shown in FIG. **10**, the heat generated in the portion where the heat generation is excessive in the battery cell **110** passes through the first heat conductive resin layer **600***a*, the bottom part **210***a* of the housing **200**, and the first heat sink **300***a* in this order to be discharged to the outside. Although not specifically shown in the figure, heat passes through the second thermal conductive resin layer **600***b*, the bottom part **210***a* of the housing **200**, and the second heat sink **300***b* in this order similarly on the opposite side for the heat dissipated to be discharged to the outside.

[0102] On the other hand, as shown in FIG. **11**, in the portion where the heat of the battery cell **110** is relatively weak, the thermal conductive resin is not applied, and thus a kind of air layer AL may be formed between the battery cell **110** and the bottom part **210***a* of the housing **200**. The air layer AL may function as a heat insulating layer and can relatively limit the heat discharge. In addition, a separate heat sink is not disposed in the corresponding part.

[0103] As described above, the degree of heat discharge is designed differently for each of the portions where heat generation of the battery cell **110** is excessive and the portion where the heat generation is minimal, thereby attempting to eliminate the temperature deviation between the respective parts of the battery cell **110**.

[0104] Considering in a comprehensive way, first heat sink **300***a* and the second heat sink **300***b* are disposed to correspond to each of the portions to which the first thermal conductive resin layer **600***a* and the second thermal conductive resin layer **600***b* are applied, whereby it is possible to maximize the cooling performance of the portion where heat generation of the battery cell stack **120** is excessive. Further, in the portion where heat generation of the battery cell stack **120** is weak, it was attempted to relatively limit the cooling performance to balance the temperature deviation.

[0105] On the other hand, referring back to FIGS. **2**, **6**, **7** and **10**, in the case of the conventional battery module **10**, the refrigerant flow path is formed inside the heat sink **60**, whereas the first heat sink **300***a* and the second heat sink **300***b* according to the present embodiment include the first recessed part **320***a* and the second recessed part **320***b* that have open upper parts, so that the bottom part **210***a* of the housing **200** is in direct contact with the refrigerant. That is, the bottom part **210***a* of the housing **200** is constituted by an upper plate of each of the first heat sink **300***a* and the second heat sink **300***b*. Thereby, the battery module **100** according to the present embodiment has an advantage that more heat can be transferred directly as compared with the conventional heat sink **60**.

[0106] Next, a battery pack according to an embodiment of the present disclosure will be described in detail with reference to FIGS. **4**, **8**, **9** and **12**.

[0107] FIG. **12** is a perspective view of a battery module, a pack refrigerant supply tube, and a pack refrigerant discharge tube according to an embodiment of the present disclosure.

[0108] Referring to FIGS. **4**, **8**, **9** and **12**, the battery pack according to an embodiment of the present disclosure includes a battery module **100**, a first pack refrigerant supply tube **1100***a*, a first pack refrigerant discharge tube **1200***a*, a second pack refrigerant supply tube **1100***b* and a second pack refrigerant discharge tube **1200***b*.

[0109] The first pack refrigerant supply tube **1100***a* supplies a refrigerant to the refrigerant flow path formed by the first heat sink **300***a*. Specifically, the first pack refrigerant supply tube **1100***a* may be connected to the refrigerant injection port **510***a*. The refrigerant that has moved through the first pack refrigerant supply tube **1100***a* passes through the refrigerant injection port **510***a* and between the first housing protrusion part **211***a* and the first heat sink protrusion part **300**Pa in this order and flows into the first heat sink **300***a*.

[0110] The first pack refrigerant discharge tube **1200***a* discharges the refrigerant from the refrigerant flow path formed by the first heat sink **300***a*. Specifically, the first pack refrigerant discharge tube **1200***a* may be connected to the refrigerant discharge port **520***a*. The refrigerant flowing into the first heat sink **300***a* passes through between the other first housing protrusion part **211***a* and the first heat sink protrusion part **300**Pa and through the refrigerant discharge port **520***a* in this order to be discharged to the first pack refrigerant discharge tube **1200***a*.

[0111] Meanwhile, the second pack refrigerant supply tube **1100***b* supplies a refrigerant to the refrigerant flow path formed by the second heat sink **300***b*. Specifically, the second pack refrigerant supply tube **1100***b* may be connected to the refrigerant injection port **510***b*. The refrigerant that has moved through the second pack refrigerant supply tube **1100***b* passes through the refrigerant injection port **510***b* and between the second housing protrusion port **211***b* and the second heat sink protrusion part **300**Pb in this order and flows into the second heat sink **300***b*.

[0112] The second pack refrigerant discharge tube **1200***b* discharges the refrigerant from the refrigerant flow path formed by the second heat sink **300***b*. Specifically, the second pack refrigerant discharge tube **1200***b* may be connected to the refrigerant discharge port **520***b*. The refrigerant flowing into the second heat sink **300***b* passes between the other second housing protrusion part **211***b* and the second heat sink protrusion part **300**Pb and through the refrigerant discharge port **520***b* in this order to be discharged to the second pack refrigerant discharge tube **1200***b*.

[0113] Next, a battery module according to another embodiment of the present disclosure will be described with reference to FIGS. **13** and **14**.

[0114] FIG. **13** is an exploded perspective view of a battery module according to another embodiment of the present disclosure. FIG. **14** is a perspective view of a lower frame and a thermal conductive resin layer included in the battery module of FIG. **13**.

[0115] Referring to FIGS. **13** and **14**, the battery module **100'** according to another embodiment of the present disclosure includes a battery cell stack **120** in which a plurality of battery cells **110** are stacked; a housing **200** that houses the battery cell stack **120**; and a first heat sink **300***a* and a second heat sink **300***b* that are located under the bottom part **210***a* of the housing **200**. The housing **200** may include a lower frame **210** and an upper cover **220**, and a first end plate **410** and a second end plate **420** may be disposed in each of the front surface and the rear surface of the battery cell stack **120**. The details concerning each configuration overlaps with the contents previously described, and thus are omitted.

[0116] At this time, the battery module **100'** according to the present embodiment may further include a thermal conductive resin layer **600***c* located between the lower surface of the battery cell stack **120** and the bottom part **210***a* of the housing **200**. A thermal conductive resin can be applied to the bottom part **210***a* of the housing **200** to form a thermal conductive resin layer **600***c*. Details concerning the thermal conductive resin overlap with the contents previously described, and thus are omitted.

[0117] The battery module **100** described above includes a first thermal conductive resin layer **600***a* and a second thermal conductive resin layer **600**b corresponding to the first heat sink **300**a and the second heat sink **300**b, respectively, whereas the battery module **100'** according to the present embodiment may include a single thermal conductive resin layer **600**c.

[0118] With respect to a direction perpendicular to one surface of the bottom part **210**a of the housing **200**, the thermal conductive resin layer **600**c according to the present embodiment covers a region where the first heat sink **300**a is located and a region where the second heat sink **300**b is located.

[0119] That is, not only can the thermal conductive resin layer **600**c cover the region where the first heat sink **300**a is located and the region where the second heat sink **300**b is located, but it can also cover the region located between them. **1**. Thereby, the heat dissipation performance in the central part of the battery cell **110** may be complemented.

[0120] The first and second thermal conductive resin layers **600**a and **600**b or a single thermal conductive resin layer **600**c may be selected and applied to improve heat dissipation performance and minimize temperature deviation for each battery module based on the type, number, size, etc. of the applied battery cells **110**.

[0121] Even though the terms indicating directions such as front, rear, left, right, upper, and lower directions are used herein, it is obvious to those skilled in the art that these merely represent for convenience of explanation, and may differ depending on a position of an observer, a position of an object, or the like.

[0122] The one or more battery modules according to the present embodiment as described above can be mounted together with various control and protection systems such as a battery management system (BMS) and a cooling system to form a battery pack.

[0123] The battery module or the battery pack can be applied to various devices. Specifically, these devices can be applied to vehicle means such as an electric bicycle, an electric vehicle, a hybrid vehicle, but the present disclosure is not limited thereto and can be applied to various devices that can use the secondary battery.

[0124] Although the preferred embodiments of the present disclosure have been described in detail above, the scope of the present disclosure is not limited thereto, and various modifications and improvements made by those skilled in the art using the basic concepts of the present disclosure defined in the following claims also fall within the spirit and scope of the present disclosure.

1. A battery module comprising:

a battery cell stack in which a plurality of battery cells including electrode leads protruding in mutually opposite directions are stacked;

a module frame that houses the battery cell stack; and

a first heat sink and a second heat sink that are located under the bottom part of the module frame;

wherein a refrigerant flow path is formed between the first heat sink and the bottom part of the module frame and between the second heat sink and the bottom part of the module frame, respectively, and

wherein the refrigerant flow path formed by the first heat sink and the refrigerant flow path formed by the second heat sink are separated from each other.

2. The battery module according to claim **1**, wherein:

the first heat sink and the second heat sink are located separately from each other along a direction parallel to a direction in which the electrode leads protrude.

3. The battery module according to claim **1**,

which further comprises a first thermal conductive resin layer and a second thermal conductive resin layer that are located between the lower surface of the battery cell stack and the bottom part of the module frame.

4. The battery module according to claim **3**, wherein:

the first thermal conductive resin layer and the second thermal conductive resin layer are located separately from each other along a direction parallel to a direction in which the electrode leads protrude.

5. The battery module according to claim **3**, wherein:

the first heat sink is located at a portion corresponding to the first thermal conductive resin layer, and the second heat sink is located at a portion corresponding to the second thermal conductive resin layer, with respect to a direction perpendicular to one surface of the bottom part of the module frame.

6. The battery module according to claim **1**, wherein:

the first heat sink comprises a first lower plate that is joined to the bottom part of the module frame, and a first recessed part that is recessed in a lower direction from the first lower plate.

7. The battery module according to claim **1**,

which further comprises a thermal conductive resin layer that is located between the lower surface of the battery cell stack and the bottom part of the module frame.

8. The battery module according to claim **7**, wherein:

the thermal conductive resin layer covers a region where the first heat sink is located and a region where the second heat sink is located, with respect to a direction perpendicular to one surface of the bottom part of the module frame.

9. The battery module according to claim **1**, wherein:

the module frame comprises first module frame protrusion part that is formed by protruding a part of the bottom part of the module frame, and

a refrigerant injection port is connected to any one of the first module frame protrusion parts and a refrigerant discharge port is connected to the other one.

10. The battery module according to claim **9**, wherein:

the first heat sink comprises first heat sink protrusion parts that protrudes from one side of the first heat sink to portions where the first module frame protrusion parts are located.

11. The battery module according to claim **1**, wherein:

the second heat sink comprises a second lower plate that is joined to the bottom part of the module frame, and a second recessed part that is recessed in a lower direction from the second lower plate.

12. The battery module according to claim **1**, wherein:

the module frame comprises second module frame protrusion parts formed by protruding a part of the bottom part of the module frame, and

a refrigerant injection port is connected to any one of the second module frame protrusion parts, and a refrigerant discharge port is connected to the other one.

13. The battery module according to claim **12**, wherein:

the second heat sink comprises second heat sink protrusion parts that protrudes from one side of the second heat sink to portions where the second module frame protrusion parts are located.

14. A battery pack comprising:

the battery module according to claim **1**;

a first pack refrigerant supply tube that supplies a refrigerant to a refrigerant flow path formed by a first heat sink;

a first pack refrigerant discharge tube that discharges the refrigerant from the refrigerant flow path formed by the first heat sink;

a second pack refrigerant supply tube that supplies a refrigerant to the refrigerant flow path formed by the second heat sink; and

a second pack refrigerant discharge tube that discharges the refrigerant from the refrigerant flow path formed by the second heat sink.

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